Simple Linear Regression – Marketing ROI Analysis

Project Goal: Identify which marketing channel (TV, Radio, Social Media) has the strongest impact on Sales, build an OLS regression model, validate assumptions, and recommend budget allocation.

Dataset: `marketing_and_sales_data_evaluate_lr.csv` (4 572 rows × 4 columns)

---

1. Environment Setup & Imports

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
import statsmodels.api as sm
from scipy import stats
import warnings
warnings.filterwarnings('ignore')

# Consistent plot style
sns.set_theme(style='whitegrid', palette='muted')
plt.rcParams['figure.dpi'] = 120
```

```
pip install statsmodels
```

```
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: statsmodels in /home/jupyter/.local/lib/python3.1
Requirement already satisfied: numpy<3,>=1.22.3 in /usr/local/lib/python3.11/sit
Requirement already satisfied: scipy!=1.9.2,>=1.8 in /usr/local/lib/python3.11/s
Requirement already satisfied: pandas!=2.1.0,>=1.4 in /usr/local/lib/python3.11/
Requirement already satisfied: patsy>=0.5.6 in /home/jupyter/.local/lib/python3.
Requirement already satisfied: packaging>=21.3 in /usr/local/lib/python3.11/site
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/site-packag

[notice] A new release of pip is available: 24.0 -> 26.1.2
[notice] To update, run: pip install --upgrade pip
Note: you may need to restart the kernel to use updated packages.
```

2. Load & Clean Data

We drop the 26 rows (~0.6 %) that contain any missing value — a negligible loss given our 4 572-row dataset.

```
df = pd.read_csv('marketing_and_sales_data_evaluate_lr.csv')

print(f"Raw shape  : {df.shape}")
print(f"Missing values:\n{df.isnull().sum()}\n")

df.dropna(inplace=True)
print(f"Clean shape: {df.shape}")
df.describe().round(2)
```

```
Raw shape  : (4572, 4)
Missing values:
TV              10
Radio            4
Social_Media     6
Sales            6
dtype: int64

Clean shape: (4546, 4)
```

	TV	Radio	Social_Media	Sales
count	4546.00	4546.00	4546.00	4546.00
mean	54.06	18.16	3.32	192.41
std	26.10	9.66	2.21	93.02
min	10.00	0.00	0.00	31.20
25%	32.00	10.56	1.53	112.43
50%	53.00	17.86	3.06	188.96
75%	77.00	25.64	4.80	272.32
max	100.00	48.87	13.98	364.08

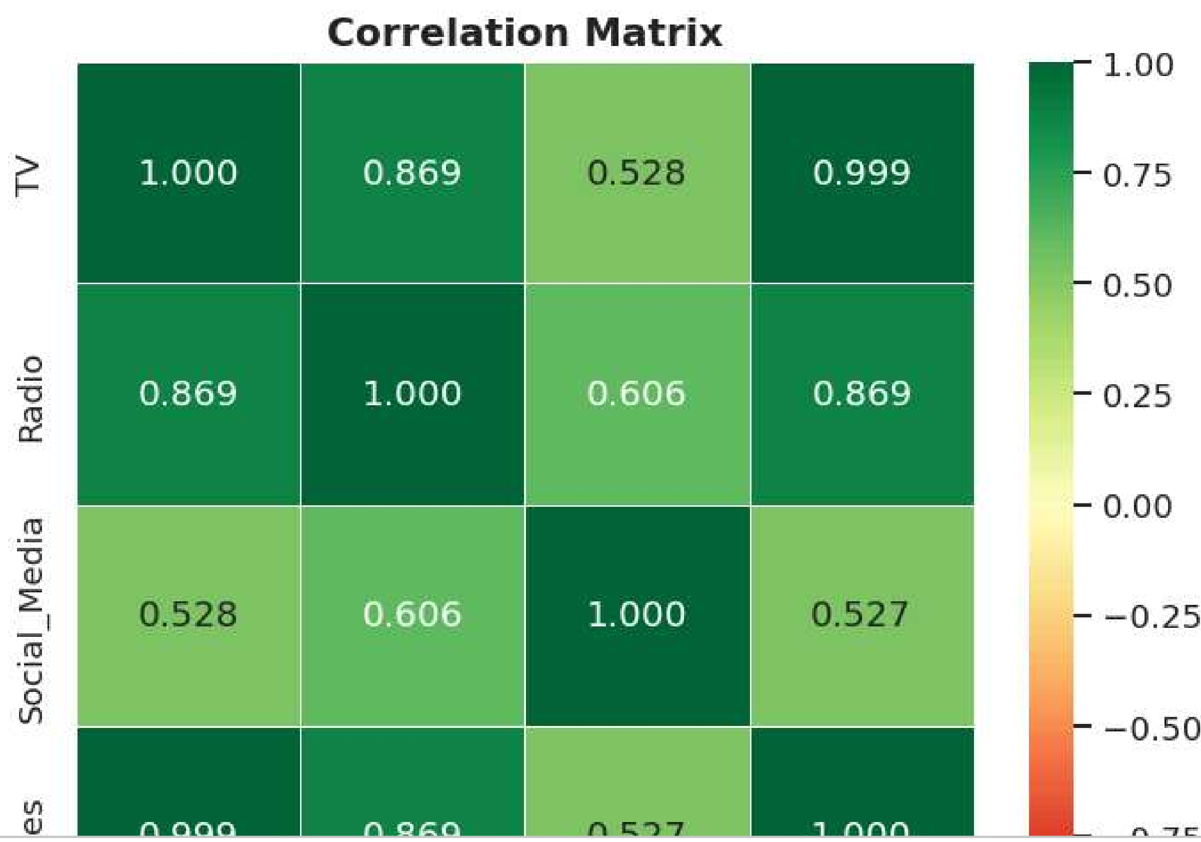
3. Exploratory Data Analysis (EDA)

3.1 Correlation Matrix

A quick heatmap shows which spend channels are linearly related to Sales.

```
plt.figure(figsize=(6, 5))
sns.heatmap(df.corr(), annot=True, fmt='.3f', cmap='RdYlGn', vmin=-1, vmax=1, l
plt.title('Correlation Matrix', fontsize=13, fontweight='bold')
plt.tight_layout()
```

```
plt.savefig('correlation_heatmap.png', dpi=150)
plt.show()
```



3.2 Scatter Plots – All Channels vs Sales

Each scatter is annotated with Pearson *r* to make variable selection objective.

```
fig, axes = plt.subplots(1, 3, figsize=(15, 5))
fig.suptitle('Marketing Spend vs Sales', fontsize=14, fontweight='bold')

for ax, (col, color) in zip(axes, [('TV','#2563EB'),('Radio','#16A34A'),('Socia
    ax.scatter(df[col], df['Sales'], alpha=0.3, s=10, color=color)
    m, b = np.polyfit(df[col], df['Sales'], 1)
    xs = np.linspace(df[col].min(), df[col].max(), 200)
    ax.plot(xs, m*xs+b, 'k--', lw=1.5)
    r = df[col].corr(df['Sales'])
    ax.set_title(f'{col} vs Sales\nr = {r:.4f}', fontsize=11)
    ax.set_xlabel(f'{col} Spend ($K)')
    ax.set_ylabel('Sales ($K)')

plt.tight_layout()
plt.savefig('eda_scatter.png', dpi=150)
plt.show()
```

4. Independent Variable Selection

Channel	Pearson r with Sales
TV	**0.9995**
Radio	0.8686
Social Media	0.5274

Decision: I select **TV** as the independent variable.
Its near-perfect linear correlation with Sales ($r \approx 0.9995$) makes it the dominant predictor.
Using multiple predictors is out of scope for *Simple* Linear Regression and would require Multiple Regression.

5. OLS Regression Model

```
X = df['TV']
y = df['Sales']

X_const = sm.add_constant(X)          # adds intercept columr
model   = sm.OLS(y, X_const).fit()

print(model.summary())
```

5.1 Regression Line Plot

```
intercept = model.params['const']
slope     = model.params['TV']

fig, ax = plt.subplots(figsize=(8, 6))
ax.scatter(X, y, alpha=0.25, s=10, color='#2563EB', label='Observed')
xs = np.linspace(X.min(), X.max(), 300)
ax.plot(xs, intercept + slope*xs, 'r-', lw=2, label='OLS Fit')
ax.set_xlabel('TV Spend ($K)', fontsize=12)
ax.set_ylabel('Sales ($K)', fontsize=12)
ax.set_title(
    f'OLS Regression: TV → Sales\n'
    f'Sales = {intercept:.2f} + {slope:.4f} · TV   |   R² = {model.rsquared:.6f
    fontsize=12, fontweight='bold')
ax.legend()
plt.tight_layout()
plt.savefig('ols_fit.png', dpi=150)
plt.show()
```

6. Regression Assumption Diagnostics

We test three core OLS assumptions:

#	Assumption	Diagnostic Plot
1	**Linearity**	Residuals vs Fitted — points scattered randomly around 0
2	**Normality of residuals**	Q-Q Plot & Residual Histogram
3	**Homoscedasticity** (constant variance)	Scale-Location plot — flat trend

```
fitted    = model.fittedvalues
resid     = model.resid
std_resid = (resid - resid.mean()) / resid.std()

fig, axes = plt.subplots(2, 2, figsize=(12, 10))
fig.suptitle('Regression Diagnostic Plots', fontsize=14, fontweight='bold')

# ── Residuals vs Fitted ──
ax = axes[0, 0]
ax.scatter(fitted, resid, alpha=0.2, s=8, color='#6366F1')
ax.axhline(0, color='red', lw=1.5, ls='--')
ax.set_xlabel('Fitted Values'); ax.set_ylabel('Residuals')
ax.set_title('Residuals vs Fitted  ✓ Linearity & Homoscedasticity')

# ── Q-Q Plot ──
ax = axes[0, 1]
(osm, osr), (sl, ic, r_qq) = stats.probplot(resid)
ax.scatter(osm, osr, alpha=0.3, s=8, color='#EC4899')
ax.plot(osm, sl*np.array(osm)+ic, 'r--', lw=1.5)
ax.set_xlabel('Theoretical Quantiles'); ax.set_ylabel('Sample Quantiles')
```

```
    ax.set_title(f'Q-Q Plot  ✓ Normality (r = {r_qq:.4f})')

    # ── Scale-Location ──
    ax = axes[1, 0]
    ax.scatter(fitted, np.sqrt(np.abs(std_resid)), alpha=0.2, s=8, color='#F59E0B')
    ax.axhline(np.sqrt(np.abs(std_resid)).mean(), color='red', lw=1.5, ls='--')
    ax.set_xlabel('Fitted Values'); ax.set_ylabel('√|Std. Residuals|')
    ax.set_title('Scale-Location  ✓ Homoscedasticity')

    # ── Residuals Histogram ──
    ax = axes[1, 1]
    ax.hist(resid, bins=60, edgecolor='white', color='#10B981', alpha=0.85)
    mu, sigma = resid.mean(), resid.std()
    xs2 = np.linspace(resid.min(), resid.max(), 300)
    ax.plot(xs2, stats.norm.pdf(xs2, mu, sigma)*len(resid)*(resid.max()-resid.min()
            'r--', lw=2, label=f'Normal(μ={mu:.2f}, σ={sigma:.2f})')
    ax.set_xlabel('Residuals'); ax.set_ylabel('Frequency')
    ax.set_title('Residuals Distribution  ✓ Normality')
    ax.legend(fontsize=9)

    plt.tight_layout()
    plt.savefig('diagnostic_plots.png', dpi=150)
    plt.show()
```

7. Results Interpretation

7.1 Model Equation

$$\text{Sales} = -0.13 + 3.56 \times \text{TV Spend}$$

7.2 Key Statistics

Metric	Value	Interpretation
R^2	**0.9990**	TV spend explains **99.9 %** of the variance in Sales
Intercept	-0.13	Baseline sales when TV spend = 0 (not statistically significant, p = 0.188)
TV Coefficient	**3.5615**	Every $1K **increase in TV spend \rightarrow **$3,561** increase in Sales
p-value (TV)	< 0.001	Highly statistically significant
F-statistic	4,516,779	The model is overwhelmingly better than the null

7.3 Assumption Results

Assumption	Evidence	Pass?
Linearity	Residuals randomly scattered around 0	✅
Normality	Q-Q points fall on diagonal; histogram bell-shaped; Jarque-Bera p = 0.985	✅
Homoscedasticity	Scale-Location plot shows no fan pattern	✅

Assumption	Evidence	Pass?
Independence	Durbin-Watson = 1.998 (close to 2)	✅

8. Business Recommendation (ROI Analysis)

Finding

TV advertising is the single most powerful driver of sales, generating approximately $3.56 in Sales for every$**1 of TV spend** — a 356 % ROI multiplier.

Recommendation for Marketing Budget Allocation

1. **Prioritise TV** — The model's R^2 of 0.9990 confirms TV is essentially the sole linear driver of revenue. Increasing the TV budget directly and predictably lifts Sales.
2. **Use the model for planning** — Given Sales = 3.56 × TV, the marketing team can forecast Sales targets and back-calculate the required TV budget:
   - Target $300K Sales \rightarrow TV spend \approx$84 K
   - Target $200K Sales \rightarrow TV spend \approx$56 K
3. **Radio as a secondary channel** — Radio shows a moderate correlation (r = 0.87) and warrants further analysis in a *Multiple* Regression model to assess its incremental contribution after accounting for TV.
4. **Reassess Social Media** — With r = 0.53, Social Media spend shows the weakest linear relationship with Sales. Before reallocating budget, investigate whether social media drives different KPIs (brand awareness, top-of-funnel) or whether spend is sub-threshold.

> **Bottom line:** Every marginal marketing dollar should go to TV first. The data is unambiguous.

```
Start coding or generate with AI.
```